(a) Recall at the fixed setting
Recall@10, new-region queries
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0σ (control)
0.5σ
1σ
2σ
Shift of the new cluster
(b) Work per query at the fixed setting
Distance computations / query
(new region, ÷ control)
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0σ (control)
0.5σ
1σ
2σ
Shift of the new cluster
(c) Speed after re-tuning to the target
QPS at target recall
(new region, ÷ control)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
× target not reached
× target not reached
× target not reached
0σ (control)
0.5σ
1σ
2σ
Shift of the new cluster
HNSW
IVF-Flat
IVF-PQ
inserted, not retrained
rebuilt on D′
(a) baseline on D; (b, c) control = 1